

RESPIRATORY DISTRESS IN NEWBORN

----- Active space -----

CLINICAL FEATURES :

1. Respiratory rate : Increased (>60 breaths/min).
 - Normal newborn respiratory rate : 40-60 breaths/min.
2. Chest retraction : D/t involvement of accessory respiratory muscles.
3. Severe hypoxia : Seen in severe cases.
 - manifest as :
 - Central cyanosis : Involves mucosa as well.
 - Grunting : Expiration against a partially closed glottis → Prevents airway/alveolar collapse.

Monitoring of Respiratory Distress

00:02:53

SCORING SYSTEMS :

1. Silverman Anderson score :

used for **preterm babies**.

	0	1	2
1. Upper chest retraction	Synchronized respiration b/w chest & abdomen	Lag : Chest lags behind abdomen during inspiration	See-saw respiration : Chest retracts, abdomen expands during expiration
2. Lower chest retraction	Not seen	mild	Severe
3. Nasal flaring	Not seen	mild	Severe
4. Xiphoid retractions	Not seen	mild	Severe
5. Grunting	Not present	Audible with stethoscope	Audible without stethoscope

Interpretation :

- <5 : mild
- 5-7 : moderate
- >7 : Severe

2. Downe's score :

- used for **term & preterm babies**.
- mnemonic for criteria : **CRARE**.

	0	1	khanirfan0392@gmail.com 2
Cyanosis	Not present	Present in room air	Present even with O ₂ support, FiO ₂ ≥ 40%
Respiratory rate	<60/min	60-80/min	>80/min
Air entry	B/L equal & normal	Decreased	Severely decreased
Retractions	Not present	mild	Severe
Grunting	Not heard	Audible with stethoscope	Audible without stethoscope

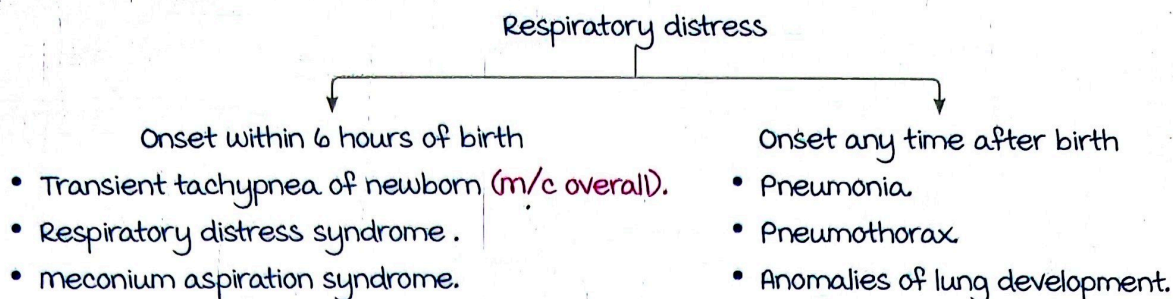
Interpretation :

- <3 : mild
- 4-6 : moderate
- >6 : Severe (Impending respiratory failure)

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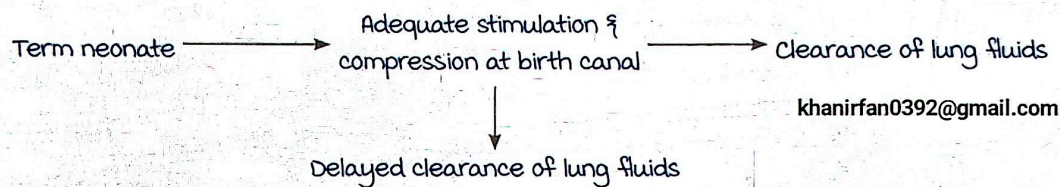
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Causes of Respiratory Distress**TRANSIENT TACHYPNOEA OF NEWBORN (TTNB) :**AKA **wet lung syndrome.**

m/c seen in :

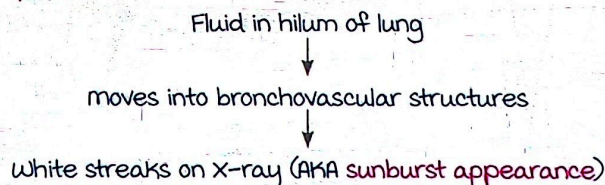
- **Term neonates.**
- Neonates delivered by lower segment caesarean section (**LSCS**).

Pathogenesis :



Chest X-ray findings :

- **Perihilar radio-opaque streaks.**



- Fluid in interlobar fissure.

management :

- Supportive management (O_2 inhalation) is sufficient as it is a transient condition.
- Improvement seen in 48-72 hours.

RESPIRATORY DISTRESS SYNDROME :AKA **hyaline membrane disease (HMD).**

m/c seen in :

- Preterm babies (<34 weeks) born to mothers who haven't received antenatal steroids (m/c : **Dexamethasone**).
- Infants of diabetic mothers (IDM).

Pathogenesis:

Preterm (<34 weeks) delivery to a mother who hasn't received steroids (Dexamethasone)

Infants of diabetic mother

↑ insulin

↓ surfactant

↑ surface tension

Alveolar collapse & respiratory distress

----- Active space -----

Note: Surfactant keeps alveoli open by decreasing surface tension on its lining.

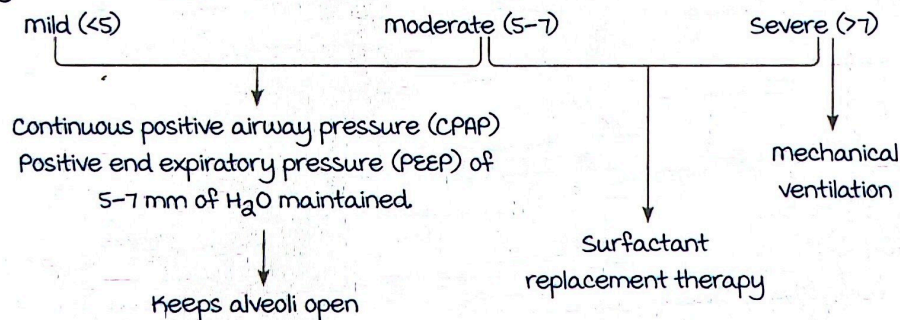
Histology: **Eosinophilic substances** in lung tissue (Resembles hyaline).

Chest X-ray findings:

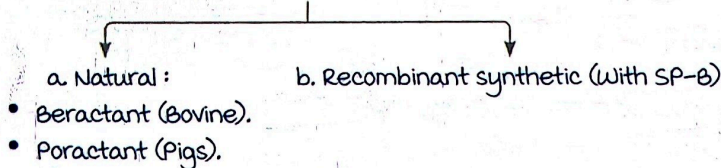
- **White out appearance**/ground glass appearance: D/t collapsed alveoli → No air entry.
- **Air bronchogram sign**: D/t open airway tract.

Management:

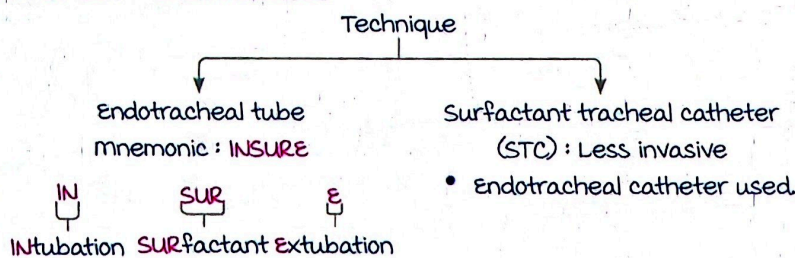
Severity assessed based on **Silverman score**:



Types of surfactant for SRT:



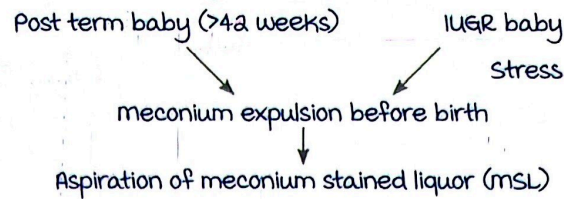
Administration of SRT: Endotracheal.



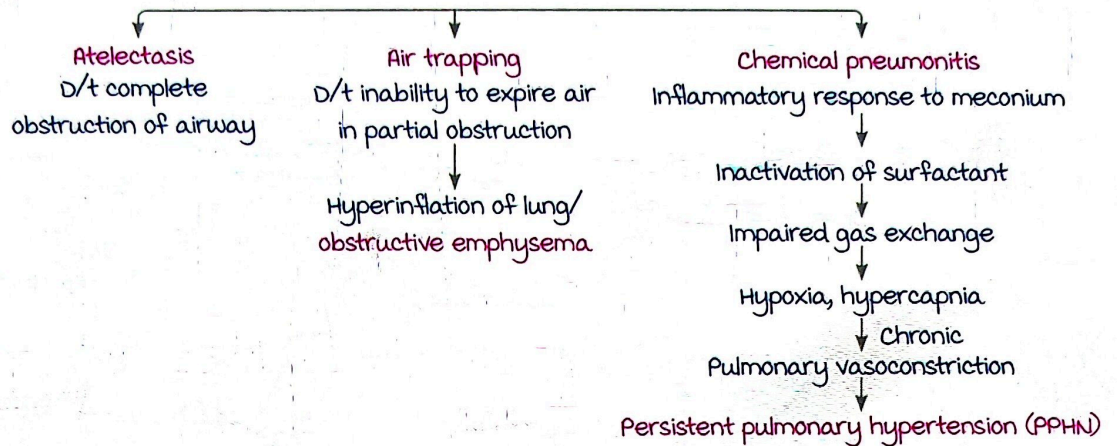
Note: Earlier synthetic surfactants leaked SP-B → ↓ efficacy.

MECONIUM ASPIRATION SYNDROME (MAS):

Pathogenesis :



Consequences of MAS :



Chest X-ray findings:

- Infiltrates in lung field: Seen in chemical pneumonitis.

Diagnosis :

- MFS : Diagnosed by exclusion and with h/o meconium-stained liquor.

management:

- Supportive management is done :
 - O₂ inhalation.
 - CPAP.
 - mechanical ventilation (Severe cases).
- Inhaled nitric oxide (iNO) : Pulmonary vasodilator to treat PPHN.



multiple small lung infiltrates

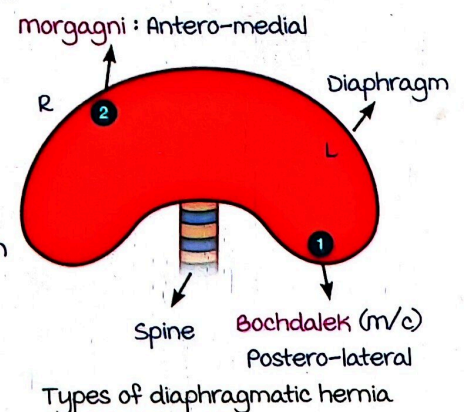
CONGENITAL DIAPHRAGMATIC HERNIA (CDH):

Pathogenesis : Anatomical defect in diaphragm.

Clinical features :

- Respiratory distress : Compression of lung in utero d/t movement of intestines into mediastinum

Pulmonary hypoplasia

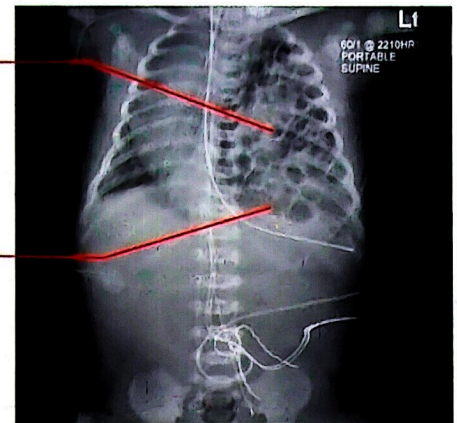


- mediastinal shift → Heart sound heard on right side.
- Sunken/scaphoid abdomen.

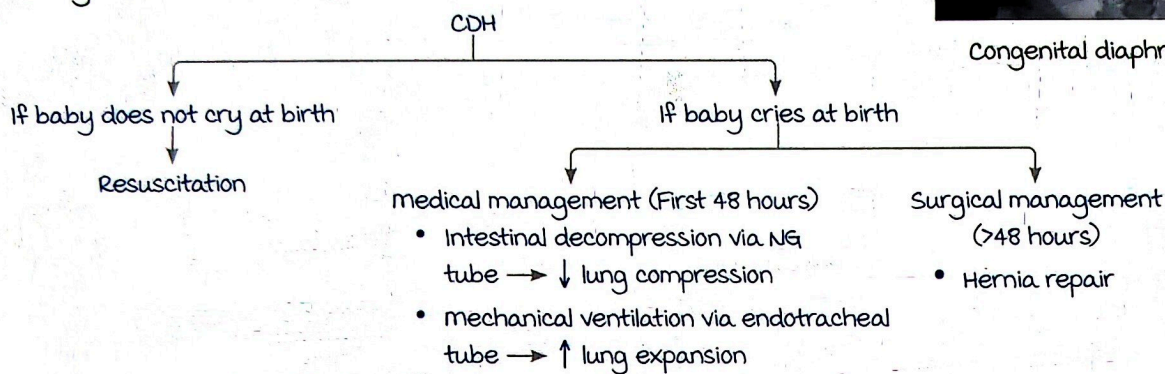
Investigations :

- USG : Antenatal.
- **Chest X-ray** : Post natal
 - Intestinal air shadows : Air-filled bubble shaped shadows d/t bowel.
 - Absent diaphragm shadows on affected side.

management :

Intestinal
air shadowAbsent
diaphragm
shadow

Congenital diaphragmatic hernia



Contraindications :

CDH is an absolute contraindication for bag & mask ventilation.

Cause : Forceful entry of air into esophagus.

↓
Inflation of bowel loops

↓
Further compression of lungs.

Prognostic factors of CDH :

Antenatal

Post natal

USG :

1. Lung-head ratio (LHR) :

- <1 : Bad prognosis.
- Unaffected lung to be considered.

2. Presence of liver in thorax : Bad prognosis.

1. PPNH (most important) : Bad prognosis.

2. Onset of respiratory distress <24 hours : Bad prognosis.